



Islamic University of Technology
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Assignment 3
Digital Forensics and Cyber Law

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Contents

1	The “Retail Ransom”	2
2	The “SpiderKash” Leak	3
3	Digital Forensics Tool	3
4	Assignment Guidelines	4
5	Marks Distribution (25 Marks)	4
6	Submission Guidelines	4

Digital Forensics and Cyber Law

1 The “Retail Ransom”

Shwapno, a major retail chain in Bangladesh, discovers that a threat actor has gained unauthorized access to its customer database, containing records for over 4 million individuals. The hackers exfiltrate the data and demand a \$1.5 million ransom to prevent its sale on the dark web. The company must decide whether to pay the ransom while forensics teams attempt to identify the “Initial Access Vector” (how they got in) and ensure the attackers haven’t left “backdoors” for a second strike.

For more context - [Daily Star](#), [The Financial Express](#)

Answer the following questions based on the given scenario

- i) How can forensics determine if the data was actually exfiltrated (stolen) or just encrypted in place?
- ii) If you were the first responder how would you act in such scenario?
- iii) If the hackers encrypted the original database, what activities (like Shadow Copy recovery or carving) could be attempted before considering the ransom?
- iv) Does paying a \$1.5M ransom constitute ‘aiding and abetting’ cyber-criminals under modern Cyber Law? What are the long-term legal risks if the company pays?
- v) Most cyber laws require ‘adequate security measures’. If forensics reveals the database was unencrypted, does the company face legal liability for the breach regardless of the hacker’s actions?

2 The “SpiderKash” Leak

On March 27, 2026, the pro-Iranian hacking collective Handala (also known as Handala Hack) announced they had successfully breached the personal Gmail account of FBI Director Kash Patel. As proof, they leaked a dossier titled “SpiderKash”, which included personal photos, his CV, and over 300 emails spanning a decade (2010–2019). The group claimed the attack was a direct retaliation for the FBI seizing four of their domains just a week prior.

For more context - [Reuters](#), [BBC](#)

Answer the following questions based on the given scenario

- i) How does a forensic investigator use DKIM (DomainKeys Identified Mail) signatures within the email headers to prove the messages actually came from Google’s servers and weren’t just typed up by the hackers in a Word document?
- ii) In the case of a Gmail breach, an investigator cannot “seize” the physical server from Google. What are the specific forensic steps used to perform a Cloud Acquisition (like using Google Takeout or API-based tools) compared to a traditional Physical Acquisition of a laptop hard drive?
- iii) If the leaked photos show Kash Patel at a specific event in 2015, how can an investigator use EXIF metadata (specifically the “Date Taken” and “GPS” tags) to verify the timeline of the hack and determine if the files were stolen from a synced cloud account or a physical backup drive?
- iv) From a forensic ethics perspective, if an investigator is cleared to look for “evidence of a hack” on a public official’s personal account, do they have the right to view all personal photos and private family emails found there? How do Cyber Laws (like the 4th Amendment in the US or similar privacy acts) help define the “boundaries” of a digital search to prevent overreach?
- v) Suppose the SpiderKash leak contains evidence of a separate financial crime. If a prosecutor wants to use those leaked emails in court, does the fact that the evidence was obtained through an illegal hack by a third party (Handala) make it inadmissible “Fruit of the Poisonous Tree”, or does the law allow it because the government itself didn’t perform the illegal hack?

3 Digital Forensics Tool

Pick one industry-standard digital forensics tool (e.g., Autopsy, FTK Imager, or Wireshark) and write a concise technical note. Your response must identify which phase of the forensic process it is used for and how it ensures the integrity of evidence for legal admissibility.

4 Assignment Guidelines

Please note that -

- You are allowed to use AI tools, but use AI responsibly.
- All sources must be properly cited.
- The report should be well-structured and professionally formatted.
- Verify all information to ensure accuracy, relevance, and factual correctness.
- For both case study, do not unnecessarily elaborate your answer, keep it relevant.

5 Marks Distribution (25 Marks)

- The Retail Ransom - 10 Marks. [1 + 1.5 + 1.5 + 3 + 3]
- The SpiderKash Leak - 12 Marks [2 + 2 + 2 + 3 + 3].
- Digital Forensics Too - 3 Marks.

6 Submission Guidelines

- Use the LaTeX template provided in the previous assignment for your report.
- Submit your report in PDF format.
- Name your file using the following format: FullID_Assignment3_HUM4747.pdf.
- Ensure your submission is made before the deadline.